1.12 Roots of Polynomials

Worked Solutions

1 a)

$\alpha + \beta = -6$ and $\alpha\beta = -7$

1 b)

$\alpha + \beta = 8$ and $\alpha\beta = 12$

1 c)

$\alpha + \beta = -\frac{3}{2}$ and $\alpha\beta = \frac{9}{2}$

1 d)

$\alpha + \beta = \frac{7}{3}$ and $\alpha\beta = -\frac{8}{3}$

1 e)

$$x + 9 = 2x^2 - 10x \Rightarrow 2x^2 - 11x - 9 = 0$$
$$x^2 - \frac{11}{2}x - \frac{9}{2} = 0$$

$\alpha + \beta = \frac{11}{2}$ and $\alpha\beta = -\frac{9}{2}$

2 a)

$$ax^2 + bx = 0 \Rightarrow x^2 + \frac{b}{a}x = 0 \Rightarrow x\left(x + \frac{b}{a}\right) = 0$$

So $\alpha = 0$ and $\beta = -\frac{b}{a}$

2 b)

$$ax^2 + c = 0 \Rightarrow x^2 + \frac{c}{a} = 0 \Rightarrow x^2 = -\frac{c}{a}$$

So $\alpha = \sqrt{-\frac{c}{a}}$ and $\beta = -\sqrt{-\frac{c}{a}}$

2 c)

$$x^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

So $\alpha = -\frac{b}{2} + \frac{\sqrt{b^2-4c}}{2}$ and $\beta = -\frac{b}{2} - \frac{\sqrt{b^2-4c}}{2}$

3 a)

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha + \beta}{\alpha\beta} \text{ as req.}$$

3 b)

Since $(\alpha+\beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$

$$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta \text{ as req.}$$

3 c)

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2}{\alpha\beta} + \frac{\beta^2}{\alpha\beta} = \frac{\alpha^2+\beta^2}{\alpha\beta}$$

Since $(\alpha+\beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} \text{ as req.}$$

3 d)

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3}{\alpha\beta} + \frac{\beta^3}{\alpha\beta} = \frac{\alpha^3+\beta^3}{\alpha\beta}$$

Since $(\alpha+\beta)^3 = \alpha^3 + 3\alpha\beta^2 + 3\alpha^2\beta + \beta^3$

$$\frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3 - 3\alpha\beta^2 - 3\alpha^2\beta}{\alpha\beta} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\alpha\beta}$$

$$\therefore \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\alpha\beta} \text{ as req.}$$

4 a)

$$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$$

$$\alpha + \beta = -\frac{5}{2}$$

4 b)

$$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$$

$$\alpha\beta = -\frac{9}{2}$$

4 c)

$$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-\frac{5}{2}}{-\frac{9}{2}} = \frac{5}{9}$$

4 d)

$$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{5}{2}\right)^2 - 2\left(-\frac{9}{2}\right) = \frac{61}{4}$$

4 e)

$$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2}{\alpha\beta} + \frac{\beta^2}{\alpha\beta} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{\left(-\frac{5}{2}\right)^2 - 2\left(-\frac{9}{2}\right)}{-\frac{9}{2}} = -\frac{61}{18}$$

4 f)

$$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3}{\alpha\beta} + \frac{\beta^3}{\alpha\beta} = \frac{\alpha^3 + \beta^3}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^3 - 3\alpha\beta^2 - 3\alpha^2\beta}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta} = \frac{\left(-\frac{5}{2}\right)^3 - 3\left(-\frac{9}{2}\right)\left(-\frac{5}{2}\right)}{-\frac{9}{2}} = \frac{395}{36}$$

5 a)

$$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$$

$$\alpha + \beta = \frac{2}{7}$$

5 b)

$$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$$

$$\alpha\beta = \frac{10}{7}$$

5 c)

$$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{\frac{2}{7}}{\frac{10}{7}} = \frac{1}{5}$$

5 d)

$$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{2}{7}\right)^2 - 2\left(\frac{10}{7}\right) = -\frac{136}{49}$$

5 e)

$$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2}{\alpha\beta} + \frac{\beta^2}{\alpha\beta} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{\left(\frac{2}{7}\right)^2 - 2\left(\frac{10}{7}\right)}{\frac{10}{7}} = -\frac{68}{35}$$

5 f)

$$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3}{\alpha\beta} + \frac{\beta^3}{\alpha\beta} = \frac{\alpha^3 + \beta^3}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^3 - 3\alpha\beta^2 - 3\alpha^2\beta}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta} = \frac{\left(\frac{2}{7}\right)^3 - 3\left(\frac{10}{7}\right)\left(\frac{2}{7}\right)}{\frac{10}{7}} = -\frac{206}{245}$$

6 a)

$$z^2 + 3z + 10 = 0$$

$\alpha + \beta = -3$ and $\alpha\beta = 10$

A quadratic with roots α^2 and β^2 requires $\alpha^2 + \beta^2$ and $\alpha^2\beta^2$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-3)^2 - 2(10) = -11$$
$$\alpha^2\beta^2 = (\alpha\beta)^2 = 10^2 = 100$$

So the equation is

$$z^2 + 11z + 100 = 0$$

6 b)

$$z^2 + 3z + 10 = 0$$

$\alpha + \beta = -3$ and $\alpha\beta = 10$

A quadratic with roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ requires $\frac{1}{\alpha}+\frac{1}{\beta}$ and $\frac{1}{\alpha\beta}$

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\beta}{\alpha\beta}+\frac{\alpha}{\alpha\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{-3}{10}=-\frac{3}{10}$$
$$\frac{1}{\alpha\beta}=\frac{1}{10}$$

So the equation is

$$z^2+\frac{3}{10}z+\frac{1}{10}=0\Rightarrow 10z^2+3z+1=0$$

7)

$\alpha+\beta=10$ and $\alpha^2+\beta^2=94$

$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\Rightarrow 94=10^2-2\alpha\beta\Rightarrow\alpha\beta=3$$
$$\therefore x^2-10x+3=0$$

So $p=10$ and $q=3$

8 a)

$$2x^2+kx-3=0\Rightarrow x^2+\frac{k}{2}x-\frac{3}{2}=0$$
$$\alpha+\beta=-\frac{k}{2}$$

8 b)

$$2x^2+kx-3=0\Rightarrow x^2+\frac{k}{2}x-\frac{3}{2}=0$$
$$\alpha\beta=-\frac{3}{2}$$

8 c)

$$2x^2+kx-3=0\Rightarrow x^2+\frac{k}{2}x-\frac{3}{2}=0$$
$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\beta}{\alpha\beta}+\frac{\alpha}{\alpha\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{-\frac{k}{2}}{-\frac{3}{2}}=\frac{k}{3}$$

8 d)

$$2x^2+kx-3=0\Rightarrow x^2+\frac{k}{2}x-\frac{3}{2}=0$$
$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\left(-\frac{k}{2}\right)^2-2\left(-\frac{3}{2}\right)=-\frac{1}{4}k^2+3$$

9)

$$3x^2 - 2x + k = 0 \Rightarrow x^2 - \frac{2}{3}x + \frac{k}{3} = 0$$

$\alpha + 3\alpha = \frac{2}{3}$ and $\alpha \times 3\alpha = \frac{k}{3}$

$\therefore \alpha = \frac{1}{6}$ and so $\frac{3}{36} = \frac{k}{3} \Rightarrow k = \frac{1}{4}$

10)

$$4x^2 - (9+k)x + k = 0 \Rightarrow x^2 - \left(\frac{9+k}{4}\right)x + \frac{k}{4} = 0$$

$\alpha + \alpha + 3 = \frac{9+k}{4}$ and $\alpha(\alpha + 3) = \frac{k}{4}$

$$\alpha + \alpha + 3 = \frac{9+k}{4} \Rightarrow 8\alpha + 12 = 9 + k \Rightarrow k = 8\alpha + 3$$

Substitute into $\alpha(\alpha + 3) = \frac{k}{4}$ gives $\alpha(\alpha + 3) = \frac{8\alpha+3}{4}$

$$4\alpha^2 + 12\alpha = 8\alpha + 3$$
$$\Rightarrow 4\alpha^2 + 4\alpha - 3 = 0$$

So $\alpha = -\frac{3}{2}$ or $\alpha = \frac{1}{2}$

So $k = 8\left(-\frac{3}{2}\right) + 3$ or $k = 8\left(\frac{1}{2}\right) + 3$

$\therefore k = -9$ or $k = 7$

11)

Let the common root be α

From $x^2 - 7x - p = 0$, substitute α in: $\alpha^2 - 7\alpha - p = 0$

From $x^2 + 9x + p = 0$, substitute α in: $\alpha^2 + 9\alpha + p = 0$

$$\alpha^2 + 9\alpha + p - (\alpha^2 - 7\alpha - p) = 0$$
$$16\alpha + 2p = 0$$
$$\therefore p = -8\alpha$$

Substituting back in:

$$\alpha^2 - 7\alpha + 8\alpha = 0 \Rightarrow \alpha^2 + \alpha = 0 \Rightarrow \alpha(\alpha + 1) = 0 \text{ so } \alpha = 0 \text{ or } \alpha = -1$$

So $p = 0$ or $p = 8$ but since p is positive, $p = 8$

12)

$$2x^2 + kx + 5 = 0 \Rightarrow x^2 + \frac{k}{2}x + \frac{5}{2} = 0$$

$\alpha + \beta = -\frac{k}{2}$ or $\alpha\beta = \frac{5}{2}$

$$(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$$
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

so that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 2\alpha\beta - 2\alpha\beta = (\alpha + \beta)^2 - 4\alpha\beta$

$$(\alpha - \beta)^2 = \left(-\frac{k}{2}\right)^2 - 4 \times \frac{5}{2} = \frac{1}{4}k^2 - 10$$

13)

$\alpha + \beta = 7$ and $\alpha\beta = 10$

$$(\alpha+\beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^3 + \beta^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha+\beta)$$
$$\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = 7^3 - 3\times 10\times 7 = 133 \text{ as req.}$$

14)

$\alpha + \beta = 4$ and $\alpha\beta = 5$

$$\begin{aligned}(\alpha+\beta)^4 &= \alpha^4 + 4\alpha^3\beta + 6\alpha^2\beta^2 + 4\alpha\beta^3 + \beta^4\\ &= \alpha^4 + \beta^4 + 6(\alpha\beta)^2 + 4\alpha\beta(\alpha^2+\beta^2)\\ &= \alpha^4 + \beta^4 + 6(\alpha\beta)^2 + 4\alpha\beta((\alpha+\beta)^2 - 2\alpha\beta)\\ \alpha^4 + \beta^4 &= (\alpha+\beta)^4 - 6(\alpha\beta)^2 - 4\alpha\beta((\alpha+\beta)^2 - 2\alpha\beta)\\ &= 4^4 - 6(5)^2 - 4\times 5(4^2 - 2\times 5) = -14 \text{ as req.}\end{aligned}$$

15)

Let the roots be α and 3α

Sum of roots: $4\alpha = -\frac{b}{a}$

Product of roots: $3\alpha^2 = \frac{c}{a}$

So that $3\left(-\frac{b}{4a}\right)^2 = \frac{c}{a} \Rightarrow \frac{3b^2}{16a^2} = \frac{c}{a} \Rightarrow 3b^2 = 16ac$ as req.

16)

Let the roots be α and $\alpha + 4$

Sum of roots: $2\alpha + 4 = -\frac{b}{a} \Rightarrow \alpha = -\frac{b}{2a} - 2$

Product of roots: $\alpha(\alpha+4) = \frac{c}{a}$

So that $\left(-\frac{b}{2a} - 2\right)\left(-\frac{b}{2a} - 2 + 4\right) = \frac{c}{a}$

$$\left(-\frac{b}{2a} - 2\right)\left(-\frac{b}{2a} + 2\right) = \frac{c}{a}$$
$$\frac{b^2}{4a^2} - 4 = \frac{c}{a}$$
$$\Rightarrow b^2 - 16a^2 = 4ac \text{ as req.}$$

17)

$\alpha + \beta = 6$ and $\alpha\beta = 8$

Sum of roots: $\alpha^3 + \beta + \alpha + \beta^3$

Product of roots: $(\alpha^3 + \beta)(\beta^3 + \alpha)$

$$\alpha^3 + \beta + \alpha + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) + \alpha + \beta = 6^3 - 3\times 8(6) + 6 = 78$$

$$\begin{aligned}(\alpha^3+\beta)(\beta^3+\alpha) &= \alpha^3\beta^3 + \beta^4 + \alpha^4 + \alpha\beta\\ &= (\alpha\beta)^3 + (\alpha+\beta)^4 - 6(\alpha\beta)^2 - 4\alpha\beta((\alpha+\beta)^2 - 2\alpha\beta) + \alpha\beta\\ &= 8^3 + 6^4 - 6\times 8^2 - 4\times 8(6^2 - 2\times 8) + 8 = 792\end{aligned}$$

$\therefore x^2 - 78x + 792 = 0$

18)

For f the sum of roots is $\frac{24}{p}$

For g the product of roots is $\frac{3}{q}$

$$\frac{24}{p} = \frac{3}{q}$$
$$\Rightarrow 8q = p$$

19)

Sum of roots: $2\beta + \beta = 3\beta = -p$

Product of roots: $2\alpha\beta = 2\beta^2 = q$

So $\beta = -\frac{p}{3}$ and so $2\left(-\frac{p}{3}\right)^2 = q$

$\therefore 2p^2 = 9q$ as req.

20)

Sum of roots: $\alpha + 5 + i = k$

Product of roots: $\alpha(5+i) = 25 + 18i$

$$\alpha = \frac{25+18i}{5+i} \times \frac{5-i}{5-i} = \frac{143+65i}{26} = \frac{11}{2} + \frac{5}{2}i$$

So $\frac{11}{2} + \frac{5}{2}i + 5 + i = k$

$\therefore k = \frac{21}{2} + \frac{7}{2}i$

21 a)

Sum of roots: $\alpha + \beta = -\frac{b}{a}$

Product of roots: $\alpha\beta = \frac{c}{a}$

New sum of roots: $\alpha + \beta + \alpha\beta = -\frac{b}{a} + \frac{c}{a}$

New product of roots: $\alpha\beta(\alpha + \beta) = \frac{c}{a} \times -\frac{b}{a}$

$$x^2 - \left(-\frac{b}{a} + \frac{c}{a}\right)x - \frac{cb}{a^2} = 0$$
$$ax^2 + a(b-c)x - bc = 0$$

21 b)

Two real distinct roots requires a positive discriminant. So:

$$(a(b-c))^2 - 4a(-bc)$$
$$= a^2(b-c)^2 + 4a^2bc$$

Since a, b and c are all positive integers

$$a^2 > 0,\ (b-c)^2 \geq 0 \text{ and } bc > 0$$

And so $a^2(b-c)^2 + 4a^2bc > 0$

Therefore the discriminant is positive and so there are two distinct real roots.

22 a)

$\alpha\beta = k$

22 b)

$\alpha + \beta = 6$

22 c)

Sum of roots: $\alpha - 2 + \beta + 2 = \alpha + \beta = 6$

Product of roots: $(\alpha - 2)(\beta + 2) = \alpha\beta + 2\alpha - 2\beta - 4$

$$\alpha\beta + 2\alpha - 2\beta - 4 = k + 2(\alpha - \beta) - 4$$

Use the solutions of $x^2 - 6x + k = 0$ to find the difference between α and β:

$$\frac{6 \pm \sqrt{36 - 4k}}{2} \text{ so } \alpha - \beta = 3 + \frac{\sqrt{36 - 4k}}{2} - \left(3 - \frac{\sqrt{36 - 4k}}{2}\right) = \sqrt{36 - 4k}$$

$\therefore k + 2(\alpha - \beta) - 4 = k + 2\sqrt{36 - 4k} - 4$

So the final quadratic is:

$$x^2 - 6x + \left(k - 4 + 2\sqrt{36 - 4k}\right) = 0 \text{ as req.}$$

23 a)

$\alpha + \beta + \gamma = -3$, $\alpha\beta + \alpha\gamma + \beta\gamma = -5$, $\alpha\beta\gamma = 1$

23 b)

$\alpha + \beta + \gamma = 9$, $\alpha\beta + \alpha\gamma + \beta\gamma = 5$, $\alpha\beta\gamma = -4$

23 c)

$$2z^3 + 5z^2 + 11z - 2 = 0 \Rightarrow z^3 + \frac{5}{2}z^2 + \frac{11}{2}z - 1 = 0$$

$\alpha + \beta + \gamma = -\frac{5}{2}$, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{11}{2}$, $\alpha\beta\gamma = 1$

23 d)

$$-5z^3 + z^2 - 12 = 0 \Rightarrow z^3 - \frac{1}{5}z^2 + \frac{12}{5} = 0$$

$\alpha + \beta + \gamma = \frac{1}{5}$, $\alpha\beta + \alpha\gamma + \beta\gamma = 0$, $\alpha\beta\gamma = -\frac{12}{5}$

23 e)

$$\begin{aligned} x + 1 &= 4x(x - 1)(x + 2) \\ x + 1 &= 4x(x^2 + x - 2) \\ x + 1 &= 4x^3 + 4x^2 - 8x \\ 4x^3 + 4x^2 - 9x - 1 &= 0 \\ \Rightarrow x^3 + x^2 - \frac{9}{4}x - \frac{1}{4} &= 0 \end{aligned}$$

$\alpha + \beta + \gamma = -1$, $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{9}{4}$, $\alpha\beta\gamma = \frac{1}{4}$

24)

$$z^3 - 3z^2 + 8z + 4 = 0$$

But the z^3 has coefficient of 2 so f$(z) = 2z^3 - 6z^2 + 16z + 8$

25 a)

Sum of roots: $2 - 1 + \alpha = -6$

So $\alpha = -7$

25 b)

Sum of pairs of products of roots: $-2 + 2\alpha - \alpha = -p$

$$-2 - 14 + 7 = -p \ \ \therefore p = 9$$

Product of roots: $-2\alpha = q \ \ \therefore q = 14$

26 a)

$$2z^3 + pz^2 - 4z + q = 0 \Rightarrow z^3 + \frac{p}{2}z^2 - 4z + \frac{q}{2} = 0$$

Sum of roots: $\alpha + \beta + \gamma = -\frac{p}{2}$ so $p = 3$

Product of roots: $\alpha\beta\gamma = -10 = -\frac{q}{2}$ so $q = 20$

26 b)

$$2z^3 + 3z^2 - 4z + 20 = 0$$
$$z \in \{-3.146, 0.823 - 1.582i, 0.823 + 1.582i\} \text{ (3 DP)}$$

27 a)

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \text{ as req.}$$

27 b)

Since $(\alpha + \beta + \gamma)^2 = (\alpha + \beta + \gamma)(\alpha + \beta + \gamma)$

$$= \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma + \alpha\gamma + \beta\gamma + \gamma^2$$
$$= \alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma$$

So that $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ as req.

27 c)

Since $(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \beta\gamma^2$$
$$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta^2 + \beta^2\gamma + \alpha\gamma^2 + \beta\gamma^2 + 3\alpha\beta\gamma$$

And

$$\begin{aligned}(\alpha+\beta)(\alpha+\gamma)(\beta+\gamma) &= (\alpha+\beta)(\alpha\beta+\alpha\gamma+\beta\gamma+\gamma^2)\\ &= \alpha^2\beta+\alpha^2\gamma+\alpha\beta\gamma+\alpha\gamma^2+\alpha\beta^2+\alpha\beta\gamma+\beta^2\gamma+\beta\gamma^2\\ &= \alpha^2\beta+\alpha^2\gamma+\alpha\gamma^2+\alpha\beta^2+\beta^2\gamma+\beta\gamma^2+2\alpha\beta\gamma\\ &= (\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma \text{ as req.}\end{aligned}$$

27 d)

Since $(\alpha\beta+\alpha\gamma+\beta\gamma)^2 = (\alpha\beta+\alpha\gamma+\beta\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)$

$$\begin{aligned}&= \alpha^2\beta^2+\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha^2\beta\gamma+\alpha^2\gamma^2+\alpha\beta\gamma^2+\alpha\beta^2\gamma+\alpha\beta\gamma^2+\beta^2\gamma^2\\ &= \alpha^2\beta^2+\beta^2\gamma^2+\alpha^2\gamma^2+2\alpha^2\beta\gamma+2\alpha\beta^2\gamma+2\alpha\beta\gamma^2\end{aligned}$$

So that

$$\begin{aligned}\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2 &= (\alpha\beta+\alpha\gamma+\beta\gamma)^2-2(\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2)\\ &= (\alpha\beta+\alpha\gamma+\beta\gamma)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)\end{aligned}$$

28 a)

$$\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha+\beta+\gamma)$$

28 b)

Since $(\alpha\beta+\alpha\gamma+\beta\gamma)^2 = (\alpha\beta+\alpha\gamma+\beta\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)$

$$\begin{aligned}&= \alpha^2\beta^2+\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha^2\beta\gamma+\alpha^2\gamma^2+\alpha\beta\gamma^2+\alpha\beta^2\gamma+\alpha\beta\gamma^2+\beta^2\gamma^2\\ &= \alpha^2\beta^2+\beta^2\gamma^2+\alpha^2\gamma^2+2\alpha^2\beta\gamma+2\alpha\beta^2\gamma+2\alpha\beta\gamma^2\end{aligned}$$

So that

$$\begin{aligned}\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2 &= (\alpha\beta+\alpha\gamma+\beta\gamma)^2-2(\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2)\\ &= (\alpha\beta+\alpha\gamma+\beta\gamma)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)\end{aligned}$$

So

$$\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2 = (\alpha\beta+\alpha\gamma+\beta\gamma)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)$$

29 a)

$$3x^3+4x^2-8x-2=0 \Rightarrow x^3+\frac{4}{3}x^2-\frac{8}{3}x-\frac{2}{3}=0$$
$$\alpha+\beta+\gamma = -\frac{4}{3}$$

29 b)

$$3x^3+4x^2-8x-2=0 \Rightarrow x^3+\frac{4}{3}x^2-\frac{8}{3}x-\frac{2}{3}=0$$
$$\alpha\beta+\alpha\gamma+\beta\gamma = -\frac{8}{3}$$

29 c)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\alpha\beta\gamma = \frac{2}{3}$$

29 d)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{-\frac{8}{3}}{\frac{2}{3}} = -4$$

29 e)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\alpha^2 + \beta^2 + \gamma^2 = \left(-\frac{4}{3}\right)^2 - 2\left(-\frac{8}{3}\right) = \frac{64}{9}$$

29 f)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \frac{2}{3}\left(-\frac{4}{3}\right) = -\frac{8}{9}$$

29 g)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\begin{aligned}\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 &= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2) \\ &= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)\end{aligned}$$

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = \left(-\frac{8}{3}\right)^2 - 2 \times \frac{2}{3}\left(-\frac{4}{3}\right) = \frac{80}{9}$$

29 h)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\begin{aligned}(\alpha+\beta)(\alpha+\gamma)(\beta+\gamma) &= (\alpha+\beta)(\alpha\beta+\alpha\gamma+\beta\gamma+\gamma^2)\\ &= \alpha^2\beta+\alpha^2\gamma+\alpha\beta\gamma+\alpha\gamma^2+\alpha\beta^2+\alpha\beta\gamma+\beta^2\gamma+\beta\gamma^2\\ &= \alpha^2\beta+\alpha^2\gamma+\alpha\gamma^2+\alpha\beta^2+\beta^2\gamma+\beta\gamma^2+2\alpha\beta\gamma\\ &= (\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma\end{aligned}$$

$$(\alpha+\beta)(\alpha+\gamma)(\beta+\gamma) = \left(-\frac{4}{3}\right)\left(-\frac{8}{3}\right) - \frac{2}{3} = \frac{26}{9}$$

29 i)

$$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$$

$$\frac{\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2}{\alpha\beta\gamma} = \frac{(\alpha\beta+\alpha\gamma+\beta\gamma)^2 - 2\alpha\beta\gamma(\alpha+\beta+\gamma)}{\alpha\beta\gamma}$$

$$\frac{\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2}{\alpha\beta\gamma} = \frac{\left(-\frac{8}{3}\right)^2 - 2\times\frac{2}{3}\left(-\frac{4}{3}\right)}{\frac{2}{3}} = \frac{40}{3}$$

30 a)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$

$$\alpha+\beta+\gamma = \frac{2}{5}$$

30 b)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$

$$\alpha\beta+\alpha\gamma+\beta\gamma = \frac{1}{5}$$

30 c)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$

$$\alpha\beta\gamma = -\frac{1}{5}$$

30 d)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$

$$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma}+\frac{\alpha\gamma}{\alpha\beta\gamma}+\frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$$

$$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} = \frac{\frac{1}{5}}{-\frac{1}{5}} = -1$$

30 e)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$
$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$
$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{2}{5}\right)^2 - 2\left(\frac{1}{5}\right) = -\frac{6}{25}$$

30 f)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$
$$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$$
$$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = -\frac{1}{5}\left(\frac{2}{5}\right) = -\frac{2}{25}$$

30 g)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$

$$\begin{aligned}\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 &= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2) \\ &= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)\end{aligned}$$

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = \left(\frac{1}{5}\right)^2 - 2 \times -\frac{1}{5}\left(\frac{2}{5}\right) = \frac{1}{5}$$

30 h)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$

$$\begin{aligned}(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) &= (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2) \\ &= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2 \\ &= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma \\ &= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma\end{aligned}$$

$$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = \left(\frac{2}{5}\right)\left(\frac{1}{5}\right) + \frac{1}{5} = \frac{7}{25}$$

30 i)

$$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$$
$$\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma} = \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{\alpha\beta\gamma}$$
$$\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma} = \frac{\left(\frac{1}{5}\right)^2 - 2 \times -\frac{1}{5}\left(\frac{2}{5}\right)}{-\frac{1}{5}} = -1$$

31)

Since there is no x^2 term $\alpha + \beta + \gamma = 0$ and so $\alpha + \beta = -\gamma$

32)

$\alpha\beta + \alpha\gamma + \beta\gamma = 0$ and $\alpha\beta\gamma = -5p$

Let $\alpha\beta = 12$ so that $12\gamma = -5p$

$\therefore \gamma = -\frac{5p}{12}$ as req.

33)

$$6x^3 + kx^2 - 26x + 15 = 0 \Rightarrow x^3 + \frac{k}{6}x^2 - \frac{13}{3}x + \frac{5}{2} = 0$$

Sum of roots: $\alpha + \beta + \gamma = \frac{11}{6}$

$$-\frac{k}{6} = \frac{11}{6} \ \therefore k = -11$$

34 a)

$$10x^3 - 57x^2 + kx - 5 = 0 \Rightarrow x^3 - \frac{57}{10}x^2 + \frac{k}{10}x - \frac{1}{2} = 0$$

Product of roots: $\alpha\beta\left(\frac{1}{\alpha}\right) = \frac{1}{2}$

So $\beta = \frac{1}{2}$

Sum of roots: $\alpha + \frac{1}{\alpha} + \beta = \frac{57}{10}$

$$\alpha + \frac{1}{\alpha} + \frac{1}{2} = \frac{57}{10}$$

$$\alpha + \frac{1}{\alpha} = \frac{52}{10}$$

$$10\alpha^2 - 52\alpha + 10 = 0 \ \therefore \alpha = 5 \text{ or } \alpha = \frac{1}{5}$$

So the solutions are 5, $\frac{1}{5}$ and $\frac{1}{2}$

34 b)

Sum of pairs of products of roots: $\alpha\left(\frac{1}{\alpha}\right) + \alpha\beta + \beta\left(\frac{1}{\alpha}\right) = \frac{k}{10}$

$$5\left(\frac{1}{5}\right) + 5 \times \frac{1}{2} + \frac{1}{2}\left(\frac{1}{5}\right) = \frac{k}{10}$$

$\therefore k = 36$

35 a)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$$\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} = 2\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)$$

Since $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$

$$\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} = 2\left(\frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}\right) = 2\left(\frac{3}{7}\right) = \frac{6}{7}$$

35 b)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$$\begin{aligned}(\alpha+3)(\beta+3)(\gamma+3) &= (\alpha+3)(\beta\gamma+3\beta+3\gamma+9)\\ &= \alpha\beta\gamma+3\alpha\beta+3\alpha\gamma+9\alpha+3\beta\gamma+9\beta+9\gamma+27\\ &= \alpha\beta\gamma+3(\alpha\beta+\alpha\gamma+\beta\gamma)+9(\alpha+\beta+\gamma)+27\end{aligned}$$

So $(\alpha+3)(\beta+3)(\gamma+3) = 7+3(3)+9(4)+27 = 79$

35 c)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$$\begin{aligned}\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2 &= (\alpha\beta+\alpha\gamma+\beta\gamma)^2 - 2(\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2)\\ &= (\alpha\beta+\alpha\gamma+\beta\gamma)^2 - 2\alpha\beta\gamma(\alpha+\beta+\gamma)\end{aligned}$$

So that

$$\frac{\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2}{5\alpha\beta\gamma} = \frac{(\alpha\beta+\alpha\gamma+\beta\gamma)^2 - 2\alpha\beta\gamma(\alpha+\beta+\gamma)}{5\alpha\beta\gamma} = \frac{3^2 - 2\times 7(4)}{5\times 7} = -\frac{47}{35}$$

35 d)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$$\begin{aligned}(\alpha+\beta)(\alpha+\gamma)(\beta+\gamma) &= (\alpha+\beta)(\alpha\beta+\alpha\gamma+\beta\gamma+\gamma^2)\\ &= \alpha^2\beta+\alpha^2\gamma+\alpha\beta\gamma+\alpha\gamma^2+\alpha\beta^2+\alpha\beta\gamma+\beta^2\gamma+\beta\gamma^2\\ &= \alpha^2\beta+\alpha^2\gamma+\alpha\gamma^2+\alpha\beta^2+\beta^2\gamma+\beta\gamma^2+2\alpha\beta\gamma\\ &= (\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma\\ (2\alpha+2\beta)(2\alpha+2\gamma)(2\beta+2\gamma) &= 8(\alpha+\beta)(\alpha+\gamma)(\beta+\gamma)\\ &= 8[(\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma]\\ &= 8[4\times 3-7] = 40\end{aligned}$$

36)

$$x^3 + kx^2 - 5x + 3 = 0$$

Sum of roots: $\alpha + \beta + \gamma = -k$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 5$

Product of roots: $\alpha\beta\gamma = -3$

$$\begin{aligned}\alpha^2\beta^2+\alpha^2\beta^2+\beta^2\gamma^2 &= (\alpha\beta+\beta\gamma+\alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha+\beta+\gamma)\\ &= 5^2 - 2(-k)(-3) = 25-6k \text{ as req.}\end{aligned}$$

37 a)

The sum of the squares of the roots is negative, which is impossible if all three roots are real because the square of any real number is non-negative and the sum of three non-negative numbers cannot be negative. So there must be one real root and two complex roots (which are complex conjugates of each other).

37 b)

$$z^3 - 7z^2 + kz - 120 = 0$$

Sum of roots: $\alpha + \beta + \gamma = 7$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = k$

Product of roots: $\alpha\beta\gamma = 120$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$
$$(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = -55$$
$$7^2 - 2k = -55$$
$$\therefore k = 52$$

38)

$$z^3 + kz + 35 - 5i = 0$$

Sum of roots: $\alpha + \beta + \gamma = 0$

Since $\alpha + \beta = 5$ then $\gamma = -5$

Product of roots: $\alpha\beta\gamma = -35 + 5i$

$$\therefore \alpha\beta = \frac{-35 + 5i}{-5} = 7 - i$$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = k$

$$\alpha\beta - 5\alpha - 5\beta = k$$
$$7 - i - 5(\alpha + \beta) = k$$
$$7 - i - 5 \times 5 = k$$
$$\therefore k = -18 - i$$

39)

$$k^3 + kz^2 + 6 + i = 0$$

Sum of roots: $\alpha + \beta + \gamma = k$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 0$

But $\alpha^2 + \beta^2 + \gamma^2 = 2i$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$
$$k^2 - 2(0) = 2i$$
$$k^2 = 2i$$

Let $k = a + bi$

$$(a + bi)^2 = 2i$$
$$a^2 + 2abi - b^2 = 2i$$

$\therefore a^2 - b^2 = 0$ and $2ab = 2 \Rightarrow a = \frac{1}{b}$

$$\left(\frac{1}{b}\right)^2 - b^2 = 0 \Rightarrow 1 - b^4 = 0$$

$b = 1$ or $b = -1$ and so $a = 1$ or $a = -1$

So $k = -1 - i$ or $k = 1 + i$

40 a)

Since $(\alpha + \beta + \gamma)^3 = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma)$

$$\begin{aligned} &= \alpha^3 + \alpha\beta^2 + \alpha\gamma^2 + 2\alpha^2\beta + 2\alpha^2\gamma + 2\alpha\beta\gamma + \alpha^2\beta + \beta^3 + \beta\gamma^2 \\ &\quad + 2\alpha\beta^2 + 2\alpha\beta\gamma + 2\beta^2\gamma + \alpha^2\gamma + \beta^2\gamma + \gamma^3 + 2\alpha\beta\gamma + 2\alpha\gamma^2 + 2\beta\gamma^2 \\ &= \alpha^3 + \beta^3 + \gamma^3 + 3\alpha\beta^2 + 3\alpha\gamma^2 + 3\alpha^2\beta + 3\alpha^2\gamma + 6\alpha\beta\gamma + 3\beta\gamma^2 + 3\beta^2\gamma \\ &= \alpha^3 + \beta^3 + \gamma^3 + 3(\alpha\beta^2 + \alpha\gamma^2 + \alpha^2\beta + \alpha^2\gamma + \beta\gamma^2 + \beta^2\gamma + 2\alpha\beta\gamma) \end{aligned}$$

But given

$$\alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma = (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$$

This gives

$$(\alpha + \beta + \gamma)^3 = \alpha^3 + \beta^3 + \gamma^3 + 3((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma)$$

So $\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)^3 - 3((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma)$

40 b)

$$2x^3 + 5x^2 - 4 = 0 \Rightarrow x^3 + \frac{5}{2}x^2 - 2 = 0$$

so $\alpha + \beta + \gamma = -\frac{5}{2}$, $\alpha\beta + \alpha\gamma + \beta\gamma = 0$ and $\alpha\beta\gamma = 2$

$$\begin{aligned} \alpha^3 + \beta^3 + \gamma^3 &= (\alpha + \beta + \gamma)^3 - 3((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma) \\ &= \left(-\frac{5}{2}\right)^3 - 3\left(\left(-\frac{5}{2}\right)(0) - 2\right) = -\frac{77}{8} \end{aligned}$$

41)

$$2x^3 + 7x^2 - 28x + 12 = 0 \Rightarrow x^3 + \frac{7}{2}x^2 - 14x + 6 = 0$$

Sum of roots: $\alpha + \beta + \gamma = -\frac{7}{2}$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = -14$

Product of roots: $\alpha\beta\gamma = -6$

Sum of new roots: $\alpha\beta + \beta\gamma + \alpha\gamma = -14$

Sum of new pairs of products of roots:

$$\begin{aligned} (\alpha\beta)(\beta\gamma) + (\alpha\beta)(\alpha\gamma) + (\beta\gamma)(\alpha\gamma) = \alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha\beta\gamma^2 &= \alpha\beta\gamma(\beta + \alpha + \gamma) \\ &= -6 \times -\frac{7}{2} = 21 \end{aligned}$$

Product of new roots: $(\alpha\beta)(\beta\gamma)(\alpha\gamma) = (\alpha\beta\gamma)^2 = (-6)^2 = 36$

So $z^3 + 14z^2 + 21z - 36 = 0$

$\therefore p = 14$, $q = 21$ and $r = -36$

42 a)

$$x^3 + px^2 + qx - 100 = 0$$

Product of roots: $\alpha\left(\frac{20}{\alpha}\right)(\alpha - 5) = 100$

$$20\alpha - 100 = 100$$
$$\therefore \alpha = 10$$

So the solutions are 10, 2 and 5

42 b)

Sum of roots: $\alpha + \frac{20}{\alpha} + \alpha - 5 = -p$

$$10 + \frac{20}{10} + 10 - 5 = -p \ \ \therefore p = -17$$

Sum of pairs of products of roots: $\alpha\left(\frac{20}{\alpha}\right) + \alpha(\alpha - 5) + \frac{20}{\alpha}(\alpha - 5) = q$

$$10\left(\frac{20}{10}\right) + 10(10 - 5) + \frac{20}{10}(10 - 5) = q \ \ \therefore q = 80$$

$p = -17$ and $q = 80$

43 a)

$$x^3 + px^2 + qx - 216 = 0$$

Product of roots: $\alpha\left(\frac{36}{\alpha}\right)(\alpha + 4) = 216$

$$\alpha + 4 = 6$$
$$\therefore \alpha = 2$$

So the solutions are 2, 18 and 6

43 b)

Sum of roots: $\alpha + \frac{36}{\alpha} + \alpha + 4 = -p$

$$2 + 18 + 6 = -p \ \ \therefore p = -26$$

Sum of pairs of products of roots: $\alpha\left(\frac{36}{\alpha}\right) + \alpha(\alpha + 4) + \frac{36}{\alpha}(\alpha + 4) = q$

$$2(18) + 2(6) + 18(6) = q \ \ \therefore q = 156$$

$p = -26$ and $q = 156$

44 a)

$$36x^3 - 444x^2 + px - q = 0 \Rightarrow x^3 - \frac{37}{3}x^2 + \frac{p}{36}x - \frac{q}{36} = 0$$

Sum of roots: $\alpha + \frac{1}{\alpha} + \alpha + \frac{1}{\alpha} = \frac{37}{3}$

$$\alpha + \frac{1}{\alpha} + \alpha + \frac{1}{\alpha} = \frac{37}{3} \Rightarrow 2\alpha + \frac{2}{\alpha} = \frac{37}{3} \Rightarrow 6\alpha^2 + 6 = 37\alpha \Rightarrow 6\alpha^2 - 37\alpha + 6 = 0$$

so $\alpha = 6$ or $\alpha = \frac{1}{6}$

$6 + \frac{1}{6} = \frac{37}{6}$ so the roots are 6, $\frac{1}{6}$ and $\frac{37}{6}$

44 b)

Sum of pairs of products of roots: $\alpha\left(\frac{1}{\alpha}\right)+\alpha\left(\alpha+\frac{1}{\alpha}\right)+\frac{1}{\alpha}\left(\alpha+\frac{1}{\alpha}\right)=\frac{p}{36}$

$$6\left(\frac{1}{6}\right)+6\left(6+\frac{1}{6}\right)+\frac{1}{6}\left(6+\frac{1}{6}\right)=\frac{1405}{36}=\frac{p}{36}\ \therefore p=1405$$

Product of roots: $\alpha\left(\frac{1}{\alpha}\right)\left(\alpha+\frac{1}{\alpha}\right)=\frac{q}{36}$

$$6\left(\frac{1}{6}\right)\left(6+\frac{1}{6}\right)=\frac{37}{6}=\frac{q}{36}\ \therefore q=222$$

So $p=1405$ and $q=222$

45 a)

$$2x^3+3x^2-7x+2=0\Rightarrow x^3+\frac{3}{2}x^2-\frac{7}{2}x+1=0$$

so $\alpha+\beta+\gamma=-\frac{3}{2}$, $\alpha\beta+\alpha\gamma+\beta\gamma=-\frac{7}{2}$ and $\alpha\beta\gamma=-1$

$$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\beta\gamma}{\alpha\beta\gamma}+\frac{\alpha\gamma}{\alpha\beta\gamma}+\frac{\alpha\beta}{\alpha\beta\gamma}=\frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}=\frac{-\frac{7}{2}}{-1}=\frac{7}{2}$$

45 b)

New roots: $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$

Sum of roots: $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{7}{2}$

Sum of pairs of products of roots: $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\alpha\gamma}=\frac{\gamma+\alpha+\beta}{\alpha\beta\gamma}=\frac{-\frac{3}{2}}{-1}=\frac{3}{2}$

Product of roots: $\frac{1}{\alpha\beta\gamma}=\frac{1}{-1}=-1$

New equation:

$$x^3-\frac{7}{2}x^2+\frac{3}{2}x+1=0\Rightarrow 2x^3-7x^2+3x+2=0$$

46 a)

$$3z^3-5z^2+7z-4=0\Rightarrow z^3-\frac{5}{3}z^2+\frac{7}{3}z-\frac{4}{3}=0$$

so $\alpha+\beta+\gamma=\frac{5}{3}$, $\alpha\beta+\alpha\gamma+\beta\gamma=\frac{7}{3}$ and $\alpha\beta\gamma=\frac{4}{3}$

New roots: $\alpha\beta$, $\alpha\gamma$ and $\beta\gamma$

Sum of roots: $\alpha\beta+\alpha\gamma+\beta\gamma=\frac{7}{3}$

Sum of pairs of products of roots:

$$(\alpha\beta)(\alpha\gamma)+(\alpha\beta)(\beta\gamma)+(\alpha\gamma)(\beta\gamma)=\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2=\alpha\beta\gamma(\alpha+\beta+\gamma)$$
$$=\frac{4}{3}\left(\frac{5}{3}\right)=\frac{20}{9}$$

Product of roots: $(\alpha\beta)(\alpha\gamma)(\beta\gamma)=(\alpha\beta\gamma)^2=\left(\frac{4}{3}\right)^2=\frac{16}{9}$

New equation:

$$z^3-\frac{7}{3}z^2+\frac{20}{9}z-\frac{16}{9}=0\Rightarrow 9z^3-21z^2+20z-16=0$$

46 b)

$$3z^3 - 5z^2 + 7z - 4 = 0 \Rightarrow z^3 - \frac{5}{3}z^2 + \frac{7}{3}z - \frac{4}{3} = 0$$

so $\alpha + \beta + \gamma = \frac{5}{3}$, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{7}{3}$ and $\alpha\beta\gamma = \frac{4}{3}$

New roots: $\alpha + \beta$, $\beta + \gamma$ and $\alpha + \gamma$

Sum of roots: $\alpha + \beta + \beta + \gamma + \alpha + \gamma = 2(\alpha + \beta + \gamma) = 2 \times \frac{5}{3} = \frac{10}{3}$

Sum of pairs of products of roots:

$$\begin{aligned}
&(\alpha + \beta)(\beta + \gamma) + (\alpha + \beta)(\alpha + \gamma) + (\beta + \gamma)(\alpha + \gamma) \\
&= \alpha\beta + \alpha\gamma + \beta^2 + \beta\gamma + \alpha^2 + \alpha\gamma + \alpha\beta + \beta\gamma + \alpha\beta + \beta\gamma + \alpha\gamma + \gamma^2 \\
&= \alpha^2 + \beta^2 + \gamma^2 + 3\alpha\beta + 3\alpha\gamma + 3\beta\gamma
\end{aligned}$$

Since $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$\begin{aligned}
&(\alpha + \beta)(\beta + \gamma) + (\alpha + \beta)(\alpha + \gamma) + (\beta + \gamma)(\alpha + \gamma) \\
&= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) + 3(\alpha\beta + \alpha\gamma + \beta\gamma) \\
&= (\alpha + \beta + \gamma)^2 + \alpha\beta + \alpha\gamma + \beta\gamma \\
&= \left(\frac{5}{3}\right)^2 + \frac{7}{3} = \frac{46}{9}
\end{aligned}$$

Product of roots:

$$\begin{aligned}
(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) &= (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2) \\
&= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2 \\
&= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma \\
&= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma \\
&= \left(\frac{5}{3}\right)\left(\frac{7}{3}\right) - \frac{4}{3} = \frac{23}{9}
\end{aligned}$$

New equation:

$$z^3 - \frac{10}{3}z^2 + \frac{46}{9}z - \frac{23}{9} = 0 \Rightarrow 9z^3 - 30z^2 + 46z - 23 = 0$$

47)

$$3x^3 + 4x^2 - 8x + k = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x + \frac{k}{3} = 0$$

so $\alpha + \beta + \gamma = -\frac{4}{3}$, $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{8}{3}$ and $\alpha\beta\gamma = -\frac{k}{3}$

$$\begin{aligned}
(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) &= (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2) \\
&= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2 \\
&= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma \\
&= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma = \left(-\frac{4}{3}\right)\left(-\frac{8}{3}\right) + \frac{k}{3} = \frac{32}{9} + \frac{k}{3}
\end{aligned}$$

48)

$$px^3 + 11x^2 - 106x + q = 0 \Rightarrow x^3 + \frac{11}{p}x^2 - \frac{106}{p}x + \frac{q}{p} = 0$$

so $\alpha + \beta + \gamma = -\frac{11}{p}$, $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{106}{p}$ and $\alpha\beta\gamma = -\frac{q}{p}$

$$20w^3 + 93w^2 - rw - 685 = 0 \Rightarrow w^3 + \frac{93}{20}w^2 - \frac{r}{20}w - \frac{137}{4} = 0$$

Sum of roots: $-\frac{93}{20}$

$$3\alpha - 1 + 3\beta - 1 + 3\gamma - 1 = -\frac{93}{20}$$
$$3(\alpha + \beta + \gamma) - 3 = -\frac{93}{20}$$
$$3\left(-\frac{11}{p}\right) - 3 = -\frac{93}{20} \Rightarrow \frac{11}{p} = \frac{11}{20} \ \therefore p = 20$$

Sum of pairs of products of roots: $-\frac{r}{20}$

$$\begin{aligned}&(3\alpha - 1)(3\beta - 1) + (3\alpha - 1)(3\gamma - 1) + (3\beta - 1)(3\gamma - 1)\\ &= 9\alpha\beta - 3\alpha - 3\beta + 1 + 9\alpha\gamma - 3\alpha - 3\gamma + 1 + 9\beta\gamma - 3\beta - 3\gamma + 1\\ &= 9\alpha\beta + 9\alpha\gamma + 9\beta\gamma - 6\alpha - 6\beta - 6\gamma + 3\\ &= 9(\alpha\beta + \alpha\gamma + \beta\gamma) - 6(\alpha + \beta + \gamma) + 3\end{aligned}$$

So that

$$9(\alpha\beta + \alpha\gamma + \beta\gamma) - 6(\alpha + \beta + \gamma) + 3 = -\frac{r}{20}$$
$$9\left(-\frac{106}{20}\right) - 6\left(-\frac{11}{20}\right) + 3 = -\frac{r}{20} \Rightarrow -\frac{207}{5} = -\frac{r}{20} \ \therefore r = 828$$

Product of roots: $\frac{137}{4}$

$$(3\alpha - 1)(3\beta - 1)(3\gamma - 1) = 27\alpha\beta\gamma - 9(\alpha\beta + \alpha\gamma + \beta\gamma) + 3(\alpha + \beta + \gamma) - 1$$

So

$$27\alpha\beta\gamma - 9(\alpha\beta + \alpha\gamma + \beta\gamma) + 3(\alpha + \beta + \gamma) - 1 = \frac{137}{4}$$
$$27\left(-\frac{q}{20}\right) - 9\left(-\frac{106}{20}\right) + 3\left(-\frac{11}{20}\right) - 1 = \frac{137}{4} \Rightarrow 27\left(-\frac{q}{20}\right) = -\frac{54}{5} \ \therefore q = 8$$

$\therefore p = 20$, $q = 8$ and $r = 828$

49 a)

$$x^3 + 10x^2 + kx - 126 = 0$$

Sum of roots: $\alpha + \beta + \gamma = -10$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = k$

Product of roots: $\alpha\beta\gamma = 94$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$
$$94 = (-10)^2 - 2(k)$$
$$\therefore k = 3$$

49 b)

Sum of roots: $2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) = -20$

Sum of pairs of products of roots: $4\alpha\beta + 4\alpha\gamma + 4\beta\gamma = 4(\alpha\beta + \alpha\gamma + \beta\gamma) = 4 \times 3 = 12$

Product of roots: $8\alpha\beta\gamma = 8 \times 126 = 1008$

$$x^3 + 20x^2 + 12x - 1008 = 0$$

50 a)

$$z^3 - pz^2 + 4z + q = 0$$

Sum of roots: $\alpha + \beta + \gamma = p$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 4$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$17 = p^2 - 2 \times 4$ so $p = \pm 5$

Since p is positive, $p = 5$

50 b)

Product of roots: $\alpha\beta\gamma = -q$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} = \frac{4}{-q}$$

$$-\frac{2}{3} = -\frac{4}{q}$$

$$\therefore q = 6$$

50 c)

$$\begin{aligned}(\alpha + 1)(\beta + 1)(\gamma + 1) &= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \alpha + \beta\gamma + \beta + \gamma + 1 \\ &= -6 + 4 + 5 + 1 = 4\end{aligned}$$

51 a)

For f the sum of roots is p

For g the product of roots is 4

$$p = 4$$

51 b)

For g:

$$3(1 - i)^3 - 12(1 - i)^2 + q(1 - i) - 12 = 0$$
$$3(-2 - 2i) - 12(-2i) + q(1 - i) - 12 = 0$$
$$-6 - 6i + 24i + q - qi - 12 = 0$$
$$-18 + q + 18i - qi = 0$$
$$\therefore q = 18$$

52 a)

$$3x^3 + kx^2 + 85x - 75 = 0 \Rightarrow x^3 + \frac{k}{3}x^2 + \frac{85}{3}x - 25 = 0$$

Product of roots: $\alpha\beta\left(\frac{\alpha}{\beta}\right) = 25$

$\alpha^2 = 25$ so $\alpha = \pm 5$

Sum of pairs of products of roots: $\alpha\beta + \alpha\left(\frac{\alpha}{\beta}\right) + \beta\left(\frac{\alpha}{\beta}\right) = \frac{85}{3}$

$$5\beta + \frac{25}{\beta} + 5 = \frac{85}{3} \Rightarrow 3\beta^2 + 15 - 14\beta = 0 \ \therefore \beta = 3 \text{ or } \beta = \frac{5}{3}$$

$$-5\beta + \frac{25}{\beta} - 5 = \frac{85}{3} \Rightarrow 3\beta^2 - 15 + 20\beta = 0 \ \therefore \beta = -\frac{10}{3} - \frac{\sqrt{145}}{3} \text{ or } \beta = -\frac{10}{3} + \frac{\sqrt{145}}{3}$$

So the possible solutions are:

5, 3 and $\frac{5}{3}$ or -5, $-\frac{10}{3} - \frac{\sqrt{145}}{3}$ and $-\frac{10}{3} + \frac{\sqrt{145}}{3}$

52 b)

Sum of roots: $-\frac{k}{3}$

$$5 + 3 + \frac{5}{3} = -\frac{k}{3} \Rightarrow k = -29$$

$$-5 - \frac{10}{3} - \frac{\sqrt{145}}{3} - \frac{10}{3} + \frac{\sqrt{145}}{3} = -\frac{k}{3} \Rightarrow k = 35$$

$k = -29$ or $k = 35$

53)

$$x^3 + kx^2 + 7 = 0$$

Sum of roots: $\alpha + \beta + \gamma = -k$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 0$

Product of roots: $\alpha\beta\gamma = -7$

$$8x^3 - 32x^2 + px + q = 0 \Rightarrow x^3 - 4x^2 + \frac{p}{8}x + \frac{q}{8} = 0$$

Sum of roots: $\alpha\beta + \gamma + \beta\gamma + \alpha + \alpha\gamma + \beta = 4$

$$-k = 4 \ \therefore k = -4$$

54)

$$z^3 - az^2 - 13z + b = 0$$

Sum of roots: $\alpha + \beta - 4 = a$

So $\alpha + \beta = a + 4$

Product of roots: $-4\alpha\beta = -b$

So $\alpha\beta = \frac{b}{4}$

Sum of pairs of products of roots: $\alpha\beta - 4\alpha - 4\beta = -13$

$$\alpha\beta - 4(\alpha+\beta) = -13 \Rightarrow \frac{b}{4} - 4(a+4) = -13$$
$$\frac{b}{4} - 4a - 16 = -13$$
$$b - 16a = 12$$
$$\therefore b = 12 + 16a$$

55 a)

$$z^3 - 4z^2 + kz - 8 = 0$$

Sum of roots: $\alpha + \beta + \alpha - \beta + \frac{8}{\alpha} = 4$

$$2\alpha + \frac{8}{\alpha} = 4 \Rightarrow 2\alpha^2 - 4\alpha + 8 = 0 \text{ so } \alpha = 1 \pm \sqrt{3}i$$

Product of roots: $\alpha\beta\left(\alpha - \beta + \frac{8}{\alpha}\right) = 8$

$$(1+\sqrt{3}i)\beta\left(1+\sqrt{3}i - \beta + \frac{8}{1+\sqrt{3}i}\right) = 8$$
$$\beta((1+\sqrt{3}i)^2 - \beta(1+\sqrt{3}i) + 8) = 8$$
$$\beta(1 + 2\sqrt{3}i - 3 - \beta - \sqrt{3}\beta i + 8) = 8$$
$$\beta + 2\sqrt{3}\beta i - 3\beta - \beta^2 - \sqrt{3}\beta^2 i + 8\beta - 8 = 0$$
$$(-\beta^2 + 6\beta - 8) + (2\sqrt{3}\beta - \sqrt{3}\beta^2)i = 0$$

Comparing real: $-\beta^2 + 6\beta - 8 = 0 \Rightarrow \beta = 2$ or $\beta = 4$

Comparing imaginary: $\beta(2\sqrt{3} - \sqrt{3}\beta) = 0 \Rightarrow \beta = 0$ or $\beta = 2$

So $\beta = 2$

Since the coefficients are real, the solutions are 2, $1 + \sqrt{3}i$ and $1 - \sqrt{3}i$

55 b)

Sum of pairs of products of roots:

$$2(1+\sqrt{3}i) + 2(1-\sqrt{3}i) + (1+\sqrt{3}i)(1-\sqrt{3}i) = k$$
$$2 + 2\sqrt{3}i + 2 - 2\sqrt{3}i + 1 + 3 = k$$
$$\therefore k = 8$$

56)

$$x^3 - (p+4)x^2 + 3px - 9p = 0$$

Sum of roots: $\alpha + \beta + \gamma = p + 4$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 3p$

Product of roots: $\alpha\beta\gamma = 9p$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha+\beta+\gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= (p+4)^2 - 2(3p) \\ &= p^2 + 2p + 16 \text{ as req.}\end{aligned}$$

57)

$$x^3 - (p+6)x^2 + (5p+4)x - (7p+3) = 0$$

Sum of roots: $\alpha + \beta + \gamma = p + 6$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 5p + 4$

Product of roots: $\alpha\beta\gamma = 7p + 3$

$$\begin{aligned}\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 &= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= (5p+4)^2 - 2(7p+3)(p+6) \\ &= 25p^2 + 40p + 16 - 2(7p^2 + 45p + 18) \\ &= 25p^2 + 40p + 16 - 14p^2 - 90p - 36 \\ &= 11p^2 - 50p - 20 \text{ as req.}\end{aligned}$$

58)

$$x^3 - (p+3)x^2 + 4px - 12p = 0$$

Sum of roots: $\alpha + \beta + \gamma = p + 3$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 4p$

Product of roots: $\alpha\beta\gamma = 12p$ $(\alpha + \beta + \gamma)^3 = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma)$

$$\begin{aligned}= \quad & \alpha^3 + \alpha\beta^2 + \alpha\gamma^2 + 2\alpha^2\beta + 2\alpha^2\gamma + 2\alpha\beta\gamma + \alpha^2\beta + \beta^3 + \beta\gamma^2 \\ & + 2\alpha\beta^2 + 2\alpha\beta\gamma + 2\beta^2\gamma + \alpha^2\gamma + \beta^2\gamma + \gamma^3 + 2\alpha\beta\gamma + 2\alpha\gamma^2 + 2\beta\gamma^2 \\ = \ & \alpha^3 + \beta^3 + \gamma^3 + 3\alpha\beta^2 + 3\alpha\gamma^2 + 3\alpha^2\beta + 3\alpha^2\gamma + 6\alpha\beta\gamma + 3\beta\gamma^2 + 3\beta^2\gamma \\ = \ & \alpha^3 + \beta^3 + \gamma^3 + 3(\alpha\beta^2 + \alpha\gamma^2 + \alpha^2\beta + \alpha^2\gamma + \beta\gamma^2 + \beta^2\gamma + 2\alpha\beta\gamma)\end{aligned}$$

But given

$$\alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma = (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$$

This gives

$$(\alpha + \beta + \gamma)^3 = \alpha^3 + \beta^3 + \gamma^3 + 3((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma)$$

$\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)^3 - 3((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma)$ And so

$$\begin{aligned}\alpha^3 + \beta^3 + \gamma^3 &= (p+3)^3 - 3((p+3)(4p) - 12p) \\ &= p^3 + 9p^2 + 27p + 27 - 3(4p^2 + 12p - 12p) \\ &= p^3 + 9p^2 + 27p + 27 - 12p^2 \\ &= p^3 - 3p^2 + 27p + 27\end{aligned}$$

59)

$$x^3 + px^2 + qx + r = 0$$

Sum of roots: $\alpha + \alpha + \alpha = -p$

$$3\alpha = -p$$

Sum of pairs of products of roots: $\alpha\alpha + \alpha\alpha + \alpha\alpha = q$

$$3\alpha^2 = q$$

Product of roots: $\alpha\alpha\alpha = -r$

$$\alpha^3 = -r$$

$\alpha = -\frac{p}{3}$ and $\alpha^2 = \frac{q}{3}$

So $-\frac{p}{3} \times \frac{q}{3} = -r$

$$p = \frac{9r}{q}$$

60)

$$x^3 - 49x^2 - 1029x + k = 0$$

Let the roots be p, pr and pr^2

Sum of roots: $p + pr + pr^2 = 49$

$$p(1 + r + r^2) = 49$$

Sum of pairs of products of roots: $p(pr) + p(pr^2) + (pr)(pr^2) = -1029$

$$p^2r + p^2r^2 + p^2r^3 = -1029$$
$$p^2r(1 + r + r^2) = -1029$$

Product of roots: $p(pr)(pr^2) = -k$

$$(pr)^3 = k$$

Substituting $p(1 + r + r^2) = 49$ into $p^2r(1 + r + r^2) = -1029$ gives

$$49pr = -1029$$
$$\therefore pr = -21$$

So $-k = (-21)^3 = -9261 \ \ \therefore k = 9261$

61)

$$ax^3 + bx^2 + cx + d = 0 \Rightarrow x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} = 0$$

Let the roots be p, pr and pr^2

Sum of roots: $p + pr + pr^2 = -\frac{b}{a}$

$$p(1 + r + r^2) = -\frac{b}{a}$$

Sum of pairs of products of roots: $p(pr) + p(pr^2) + (pr)(pr^2) = \frac{c}{a}$

$$p^2r + p^2r^2 + p^2r^3 = \frac{c}{a}$$
$$p^2r(1 + r + r^2) = \frac{c}{a}$$

Product of roots: $p(pr)(pr^2) = -\frac{d}{a}$

$$p^3r^3 = -\frac{d}{a}$$

Substituting $p(1 + r + r^2) = -\frac{b}{a}$ into $p^2r(1 + r + r^2) = \frac{c}{a}$ gives:

$$pr\left(-\frac{b}{a}\right) = \frac{c}{a}$$

Cubing both sides gives:

$$(pr)^3\left(-\frac{b}{a}\right)^3=\left(\frac{c}{a}\right)^3$$
$$\left(-\frac{d}{a}\right)\left(-\frac{b}{a}\right)^3=\left(\frac{c}{a}\right)^3$$
$$\frac{b^3d}{a^4}=\frac{c^3}{a^3}$$
$$\Rightarrow b^3d=ac^3\Rightarrow ac^3-db^3=0 \text{ as req.}$$

62)

$$2x^3+6x^2-26x-k=0\Rightarrow x^3+3x^2-13x-\frac{k}{2}=0$$

Let the roots be $a-d$, a and $a+d$

Sum of roots: $a-d+a+a+d=-3$

$$3a=-3$$

So $a=-1$

Sum of pairs of products of roots: $-(a-d)+a(a+d)+(a-d)(a+d)=-13$

$$-(-1-d)-(-1+d)+(-1-d)(-1+d)=-13$$
$$1+d+1-d+1-d^2=-13$$
$$-d^2+3=-13$$
$$d^2=16$$

So $d=\pm4$

Product of roots: $(a-d)a(a+d)=\frac{k}{2}$

$$-1(-1-4)(-1+4)=\frac{k}{2}$$
$$15=\frac{k}{2}$$
$$\therefore k=30$$

63)

$$px^3+qx^2+rx+s=0\Rightarrow x^3+\frac{q}{p}x^2+\frac{r}{p}x+\frac{s}{p}=0$$

Let the roots be $\alpha-d$, α and $\alpha+d$

Sum of roots: $\alpha-d+\alpha+\alpha+d=-\frac{q}{p}$

$$3\alpha=-\frac{q}{p}$$

Sum of pairs of products of roots: $\alpha(\alpha-d)+\alpha(\alpha+d)+(\alpha-d)(\alpha+d)=\frac{r}{p}$

$$\alpha^2-d\alpha+\alpha^2+d\alpha+\alpha^2-d^2=\frac{r}{p}$$
$$2\alpha^2+\alpha^2-d^2=\frac{r}{p}$$

Product of roots: $\alpha(\alpha - d)(\alpha + d) = -\frac{s}{p}$

$$\alpha(\alpha^2 - d^2) = -\frac{s}{p}$$

Substituting $\alpha(\alpha^2 - d^2) = -\frac{s}{p}$ into $2\alpha^2 + \alpha^2 - d^2 = \frac{r}{p}$ gives:

$$2\alpha^2 - \frac{s}{p\alpha} = \frac{r}{p}$$

Since $\alpha = -\frac{q}{3p}$

$$2\left(-\frac{q}{3p}\right)^2 - \frac{s}{p\left(-\frac{q}{3p}\right)} = \frac{r}{p}$$
$$\frac{2q^2}{9p^2} + \frac{3s}{q} = \frac{r}{p}$$
$$2q^3 + 27p^2 s = 9pqr \text{ as req.}$$

64)

$$x^3 + px^2 + qx + r = 0$$

Sum of roots: $\alpha + \beta + \gamma = -p$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = q$

Product of roots: $\alpha\beta\gamma = -r$

$$q^2 - 2pr = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

But since

$$\begin{aligned}(\alpha\beta + \alpha\gamma + \beta\gamma)^2 &= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 + 2\alpha^2\beta\gamma + 2\alpha\beta^2\gamma + 2\alpha\beta\gamma^2 \\ &= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 - 2(\alpha + \beta + \gamma)(\alpha\beta\gamma)\end{aligned}$$

So that

$$\begin{aligned}q^2 - 2pr &= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 + 2(\alpha + \beta + \gamma)(\alpha\beta\gamma) - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 \geq 0\end{aligned}$$

$\Rightarrow q^2 \geq 2pr$ as req.

65 a)

$\sum \alpha = -5$, $\sum \alpha\beta = -3$, $\sum \alpha\beta\gamma = -6$ and $\sum \alpha\beta\gamma\delta = -2$

65 b)

$\sum \alpha = 4$, $\sum \alpha\beta = 4$, $\sum \alpha\beta\gamma = 1$ and $\sum \alpha\beta\gamma\delta = 8$

65 c)

$\sum \alpha = -\frac{5}{2}$, $\sum \alpha\beta = 3$, $\sum \alpha\beta\gamma = -\frac{11}{2}$ and $\sum \alpha\beta\gamma\delta = \frac{1}{2}$

65 d)

$\sum \alpha = -1$, $\sum \alpha\beta = -\frac{7}{4}$, $\sum \alpha\beta\gamma = 5$ and $\sum \alpha\beta\gamma\delta = \frac{1}{2}$

65 e)

$$x^4 + 7 - x^3 + 4x^2 = 0$$

$\sum \alpha = 1$, $\sum \alpha\beta = 4$, $\sum \alpha\beta\gamma = 0$ and $\sum \alpha\beta\gamma\delta = 7$

66 a)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\alpha + \beta + \gamma + \delta = -\frac{1}{8}$$

66 b)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = -\frac{3}{8}$$

66 c)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = -\frac{5}{8}$$

66 d)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\alpha\beta\gamma\delta = -\frac{7}{8}$$

66 e)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\begin{aligned}\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} &= \frac{\beta\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\gamma}{\alpha\beta\gamma\delta} \\ &= \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{-\frac{5}{8}}{-\frac{7}{8}} = \frac{5}{7}\end{aligned}$$

66 f)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 + \delta^2 &= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= \left(-\frac{1}{8}\right)^2 - 2\left(-\frac{3}{8}\right) = \frac{49}{64}\end{aligned}$$

66 g)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\alpha^2\beta^2\gamma\delta + \alpha^2\beta\gamma^2\delta + \alpha^2\beta\gamma\delta^2 + \alpha\beta^2\gamma^2\delta + \alpha\beta^2\gamma\delta^2 + \alpha\beta\gamma^2\delta^2$$

$$\begin{aligned}&= (\alpha\beta\gamma\delta)(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= \left(-\frac{7}{8}\right)\left(-\frac{3}{8}\right) = \frac{21}{64}\end{aligned}$$

66 h)

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$$

$$\begin{aligned}\frac{2}{\alpha\beta\gamma} + \frac{2}{\alpha\beta\delta} + \frac{2}{\alpha\gamma\delta} + \frac{2}{\beta\gamma\delta} &= 2\left[\frac{\delta}{\alpha\beta\gamma\delta} + \frac{\gamma}{\alpha\beta\gamma\delta} + \frac{\beta}{\alpha\beta\gamma\delta} + \frac{\alpha}{\alpha\beta\gamma\delta}\right] \\ &= 2\left(\frac{\alpha + \beta + \gamma + \delta}{\alpha\beta\gamma\delta}\right) = 2\left(\frac{-\frac{1}{8}}{-\frac{7}{8}}\right) = \frac{2}{7}\end{aligned}$$

67 a)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\alpha + \beta + \gamma + \delta = -\frac{1}{4}$$

67 b)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = -\frac{1}{6}$$

67 c)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = \frac{1}{3}$$

67 d)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\alpha\beta\gamma\delta = \frac{1}{4}$$

67 e)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\begin{aligned}\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} &= \frac{\beta\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\gamma}{\alpha\beta\gamma\delta} \\ &= \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{\frac{1}{3}}{\frac{1}{4}} = \frac{4}{3}\end{aligned}$$

67 f)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 + \delta^2 &= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= \left(-\frac{1}{4}\right)^2 - 2\left(-\frac{1}{6}\right) = \frac{19}{48}\end{aligned}$$

67 g)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\alpha^2\beta^2\gamma\delta + \alpha^2\beta\gamma^2\delta + \alpha^2\beta\gamma\delta^2 + \alpha\beta^2\gamma^2\delta + \alpha\beta^2\gamma\delta^2 + \alpha\beta\gamma^2\delta^2$$

$$\begin{aligned}&= (\alpha\beta\gamma\delta)(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= \left(\frac{1}{4}\right)\left(-\frac{1}{6}\right) = -\frac{1}{24}\end{aligned}$$

67 h)

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$$

$$\begin{aligned}\frac{2}{\alpha\beta\gamma} + \frac{2}{\alpha\beta\delta} + \frac{2}{\alpha\gamma\delta} + \frac{2}{\beta\gamma\delta} &= 2\left[\frac{\delta}{\alpha\beta\gamma\delta} + \frac{\gamma}{\alpha\beta\gamma\delta} + \frac{\beta}{\alpha\beta\gamma\delta} + \frac{\alpha}{\alpha\beta\gamma\delta}\right] \\ &= 2\left(\frac{\alpha + \beta + \gamma + \delta}{\alpha\beta\gamma\delta}\right) = 2\left(\frac{-\frac{1}{4}}{\frac{1}{4}}\right) = -2\end{aligned}$$

68 a)

$$2z^4 + 3z^3 - 2z + 3 = 0 \Rightarrow z^4 + \frac{3}{2}z^3 - z + \frac{3}{2} = 0$$

$$\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{3}{2}$$
$$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 0$$
$$\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = 1$$
$$\sum \alpha\beta\gamma\delta = \alpha\beta\gamma\delta = \frac{3}{2}$$

$$\begin{aligned}\frac{3}{\alpha} + \frac{3}{\beta} + \frac{3}{\gamma} + \frac{3}{\delta} &= 3\left[\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}\right] \\ &= 3\left[\frac{\beta\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\gamma}{\alpha\beta\gamma\delta}\right] \\ &= 3\left[\frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta}\right] \\ &= 3\left[\frac{1}{\frac{3}{2}}\right] = 2\end{aligned}$$

68 b)

$$2z^4 + 3z^3 - 2z + 3 = 0 \Rightarrow z^4 + \frac{3}{2}z^3 - z + \frac{3}{2} = 0$$

$$\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{3}{2}$$
$$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 0$$
$$\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = 1$$
$$\sum \alpha\beta\gamma\delta = \alpha\beta\gamma\delta = \frac{3}{2}$$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 + \delta^2 &= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= \left(-\frac{3}{2}\right)^2 - 2(0) = \frac{9}{4}\end{aligned}$$

68 c)

$$\begin{aligned}&(1-\alpha)(1-\beta)(1-\gamma)(1-\delta) \\ &= \alpha\beta\gamma\delta - (\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta) \\ &\quad + (\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) - (\alpha + \beta + \gamma + \delta) + 1 \\ &= \frac{3}{2} - 1 - 0 - \left(-\frac{3}{2}\right) + 1 = 3\end{aligned}$$

69 a)

$$6x^4 - 35x^3 + 62x^2 - px + q = 0 \Rightarrow x^4 - \frac{35}{6}x^3 + \frac{31}{3}x^2 - \frac{p}{6}x + \frac{q}{6} = 0$$

$$\sum \alpha\beta\gamma\delta = \alpha \times \frac{1}{\alpha} \times \beta \times \frac{1}{\beta} = 1 = \frac{q}{6}$$

$\therefore q = 6$

69 b)

$\sum \alpha = \frac{35}{6}$, $\sum \alpha\beta\gamma = \frac{p}{6}$

Since

$$\sum \alpha = \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta} = \frac{35}{6}$$

$$\sum \alpha\beta\gamma = \alpha\beta\left(\frac{1}{\alpha}\right) + \alpha\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) + \alpha\beta\left(\frac{1}{\beta}\right) + \beta\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) = \beta + \frac{1}{\beta} + \alpha + \frac{1}{\alpha} = \frac{p}{6}$$

$\therefore p = 35$

70 a)

$$2x^4 + 3x^3 + px^2 + qx + 882 = 0 \Rightarrow x^4 + \frac{3}{2}x^3 + \frac{p}{2}x^2 + \frac{q}{2}x + 441 = 0$$

$\sum \alpha = -\frac{3}{2}$, $\sum \alpha\beta = \frac{p}{2}$, $\sum \alpha\beta\gamma = -\frac{q}{2}$, $\sum \alpha\beta\gamma\delta = 441$

So

$$\alpha + 2\alpha + \beta + \frac{\beta}{2} = -\frac{3}{2} \Rightarrow 3\alpha + \frac{3}{2}\beta = -\frac{3}{2} \Rightarrow 2\alpha + \beta = -1 \Rightarrow \beta = -1 - 2\alpha$$

$$\alpha \times 2\alpha \times \beta \times \frac{\beta}{2} = \alpha^2\beta^2 = 441$$

Solving simultaneously:

$$\alpha^2(-1 - 2\alpha)^2 = 441$$
$$\alpha^2(4\alpha^2 + 4\alpha + 1) = 441$$
$$4\alpha^4 + 4\alpha^3 + \alpha^2 - 441 = 0$$
$$(\alpha - 3)(2\alpha + 7)(2\alpha^2 + \alpha + 21) = 0$$

Real solutions are: $\alpha = -\frac{7}{2}$ or $\alpha = 3$

$\alpha = -\frac{7}{2}$, $\beta = 6$ or $\alpha = 3$, $\beta = -7$

So the roots are $-\frac{7}{2}$, -7, 6 and 3

70 b)

$$(2x + 7)(x - 6)(x - 3)(x + 7) = 0$$
$$(2x^2 - 5x - 42)(x^2 + 4x - 21) = 0$$
$$2x^4 + 8x^3 - 42x^2 - 5x^3 - 20x^2 + 105x - 42x^2 - 168x + 882 = 0$$
$$2x^4 + 3x^3 - 104x^2 - 63x + 882 = 0$$

$\therefore p = -104$ and $q = -63$

71 a)

$$9x^4 - 150x^3 + px^2 - qx + 63 = 0 \Rightarrow x^4 - \frac{150}{9}x^3 + \frac{p}{9}x^2 - \frac{q}{9}x + 7 = 0$$

Let $\delta = \gamma$ so that the roots are α, β, γ and γ

And $\alpha\beta = 63$

$\alpha\beta\gamma^2 = 7$ so $63\gamma^2 = 7$ so either $\gamma = -\frac{1}{3}$ or $\gamma = \frac{1}{3}$

$$\alpha + \beta + 2\gamma = \frac{150}{9} \Rightarrow \alpha + \beta = 16 \Rightarrow \beta = 16 - \alpha$$

Solving simultaneously gives:

$$\begin{gathered}\alpha(16 - \alpha) = 63 \\ \alpha^2 - 16\alpha + 63 = 0 \\ (\alpha - 7)(\alpha - 9) = 0 \\ \alpha = 7 \text{ or } \beta = 9\end{gathered}$$

So the solutions are 7, 9, $\frac{1}{3}$ and $\frac{1}{3}$

71 b)

$$\begin{aligned}&(x-7)(x-9)(3x-1)(3x-1) \\ &= (x^2 - 16x + 63)(9x^2 - 6x + 1) \\ &= 9x^4 - 6x^3 + x^2 - 144x^3 + 96x^2 - 16x + 567x^2 - 378x + 63 \\ &= 9x^4 - 150x^3 + 664x^2 - 394x + 63\end{aligned}$$

$\therefore p = 664$ and $q = 394$

72)

Roots are: α, $\beta = k\alpha$, $\gamma = k^2\alpha$ and $\delta = k^3\alpha$

$$\begin{aligned}c &= \alpha(k\alpha) + \alpha(k^2\alpha) + \alpha(k^3\alpha) + k\alpha(k^2\alpha) + k\alpha(k^3\alpha) + k^2\alpha(k^3\alpha) \\ &= \alpha^2[k + k^2 + k^3 + k^3 + k^4 + k^5] \\ &= \alpha^2\left[k^3 + \sum_{r=1}^{5} k^r\right]\end{aligned}$$

So $n = 5$